

Supplementary Materials

**Revisiting the Syngas Conversion to Olefins
over Fe-Mn Bimetallic System: Insights
from the Promxity Effects**

Ding, Xiaoxu; Zhu, Minghui; Han, Yifan;Yang, Zixu**

*State Key Laboratory of Chemical Engineering, East China University of Science and
Technology, Shanghai 200237, China*

* Corresponding author phone: (+86)21-64251928; fax: (86)21-64251928;

E-mail: yifanhan@ecust.edu.cn ; zixu.yang@ecust.edu.cn

DFT computational models and methods

The calculated Fe_3O_4 is a cubic crystal with an inverse spinel structure and the FeO is a cubic crystal adopting a rocksalt structure. Their optimized unit cells show a lattice constant of 8.381 Å and 4.308 Å, respectively, which agree well with the experimental value. Surface reactions were calculated on a p (1 × 2) supercell of Fe_3O_4 (311) surface and a p (2 × 3) supercell of FeO (200) surface. To match the experimental $\text{Mn}_{0.43}\text{Fe}_{2.57}\text{O}_4$, 4 Fe atoms in Fe_3O_4 supercell were replaced by Mn. In order to account for the lattice expansion caused by Mn, a FeO single cell with two Fe atoms replaced by Mn was individually optimized, and the lattice constant is 4.401 Å. The supercell model, similar to that of FeO (200) slab, was developed and is used to simulate the experimental $\text{Fe}_{0.497}\text{Mn}_{0.503}\text{O}$ phase. To eliminate the interaction between the repeating slabs, the vacuum gap spacing for all models were set to 15 Å. The calculated pure or Mn-substituted Fe_3O_4 (311) model has 24 Fe or Mn atoms and 32 O atoms, with 12 Fe and 16 O atoms at the bottom fixed, whereas the pure or Mn-substituted FeO (200) model has 36 Fe or Mn atoms and 36 O atoms, with the bottom 24 atoms fixed. The top and side views of the calculated slab models are shown in Fig S3.

All calculations were performed by using the Vienna Ab initio Simulation Package (VASP) with the projected augmented wave (PAW) pseudopotentials. The electron exchange and correlation energies were treated with the Perdew-Burke-Ernzerhof (PBE) functional within the generalized gradient approximation (GGA) with a kinetic energy cutoff of 400 eV [1]. DFT + U using the simplified rotationally invariant approach introduced by Dudarev et al. [2], with $U_{\text{eff}} = 3.8$ eV for Fe and $U_{\text{eff}} = 4.5$ eV for Mn were chosen to describe the 3d states of Fe and Mn [3, 4]. A Gaussian smearing with sigma = 0.2 eV was employed to converge the electronic self-consistent cycles. K-point grids of $4 \times 4 \times 4$ and $7 \times 7 \times 7$ for the Fe_3O_4 and FeO single crystals, and $3 \times 2 \times 1$ for the slab models were used to sample the Brillouin zones. The force criterion for geometry optimization was set to be < 0.05 eV/Å.

The formation energy of a surface oxygen vacancy by reacting with H_2 is calculated via eq 1:

$$\Delta E_{ov} = E_{slab-ov} + E_{H_2O} - (E_{slab} + E_{H_2}) \quad (1)$$

Where E_{slab} and $E_{slab-ov}$ are the total energies of the slabs without and with an oxygen vacancy, E_{H_2} and E_{H_2O} are the total energies of the isolated H_2 and H_2O molecules.

The penetration energy of a C atom from CO with O reacting with H_2 is defined in eq 2:

$$\Delta E_C = E_{slab-C} + E_{H_2O} - (E_{slab-ov} + E_{H_2} + E_{CO}) \quad (2)$$

Where E_{slab-C} is the total energy of the slab with carbon atom embedded into the oxygen vacancy, E_{CO} is the total energy of the isolated CO molecule.

As such, the carburization of Fe oxides can be seen as the combination of the above two processes, so that the total reaction energy is calculated as the sum of ΔE_{ov}

and ΔE_C :

$$\Delta E_{total} = \Delta E_{ov} + \Delta E_C = E_{slab-C} + 2E_{H_2O} - (E_{slab} + 2E_{H_2} + E_{CO}) \quad (3)$$

Table S1. Residual contents of C and H in Fe4Mn1-SCM catalysts with different combustion time.

Time/h	C/%	H/%
0.5	0.315	0.478
2	0.11	0.346
4	0.11	0.344

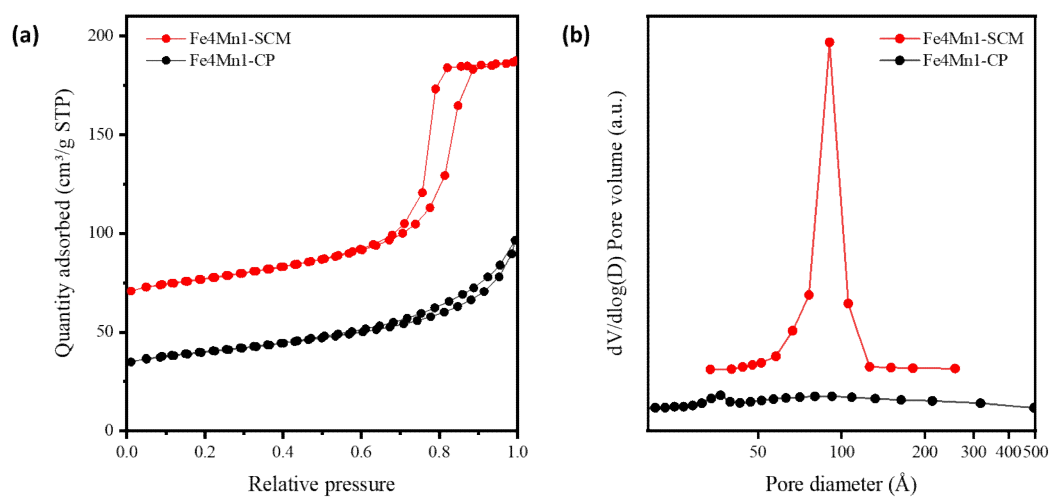


Fig S1. (a) Adsorption-desorption isotherms of different Fe4Mn1 catalysts. (b) Pore size distribution of different Fe4Mn1 catalysts.

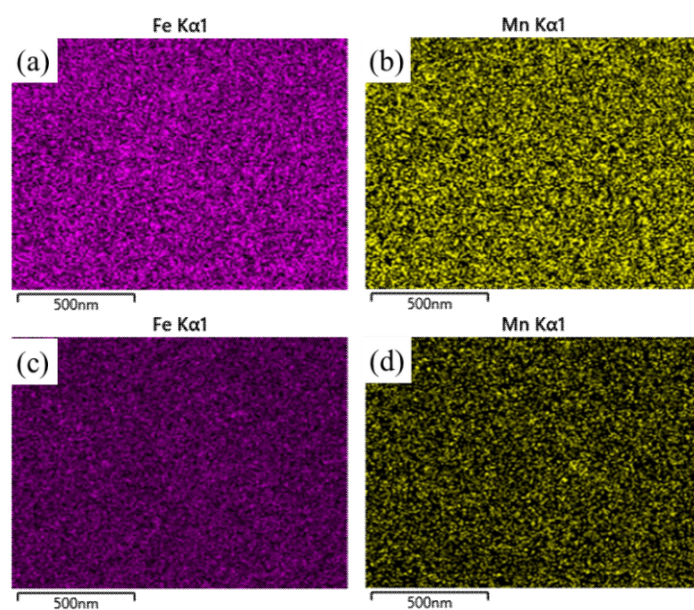


Fig S2. Elemental mapping of Fe₄Mn₁-SCM (a, b) and Fe₄Mn₁-CP (c, d).

Table S2. Detailed MES results of activated Fe4Mn1 catalysts.

	Hhf / KOe	IS / mm·s ⁻¹	QS / mm·s ⁻¹	Area	Phases
Fe4Mn1-CP	—	0.21	0.95	0.021	Fe ³⁺
	—	1.11	0.96	0.056	Fe ²⁺
	210.88	0.26	-0.08	0.36	
	184.55	0.22	0	0.432	Fe ₅ C ₂
	110	0.2	-0.07	0.132	
	—	0.41	0.8	0.141	Fe ³⁺
Fe4Mn1-SCM	—	0.97	0.78	0.183	Fe ²⁺
	215.89	0.24	-0.08	0.251	
	183.23	0.22	-0.04	0.238	Fe ₅ C ₂
	102.96	0.23	0.37	0.138	
	201.17	0.22	-0.04	0.09	Fe ₃ C

Table S3. Detailed FFT analysis results of spent Fe4Mn1 catalysts.

Phase			R1 / Å	R2 / Å	R3 / Å	<R1,R2>	<R1,R3>	<R2,R3>
Fe4Mn1-S CM	Fe ₅ C ₂	FFT	2.074	1.679	1.771	117.858°	68.837°	49.021°
	89-8968	PDF	2.083	1.643	1.773	118.974°	70.821°	48.153°
	Mn _{0.43} Fe _{2.57} O ₄	FFT	2.985	3.032	2.125	89.885°	44.496°	45.389°
	89-2807	PDF	2.988	2.988	2.113	90°	45°	45°
Fe4Mn1-C P	Fe ₅ C ₂	FFT	2.025	4.261	1.977	100.713°	27.124°	73.589°
	89-8968	PDF	2.028	4.253	2.012	102.787°	27.471°	75.316°
	Mn ₅ O ₈	FFT	3.636	1.985	2.022	107.829°	75.864°	31.965°
	72-1427	PDF	3.679	1.954	2.035	109.768°	78.407°	31.361°

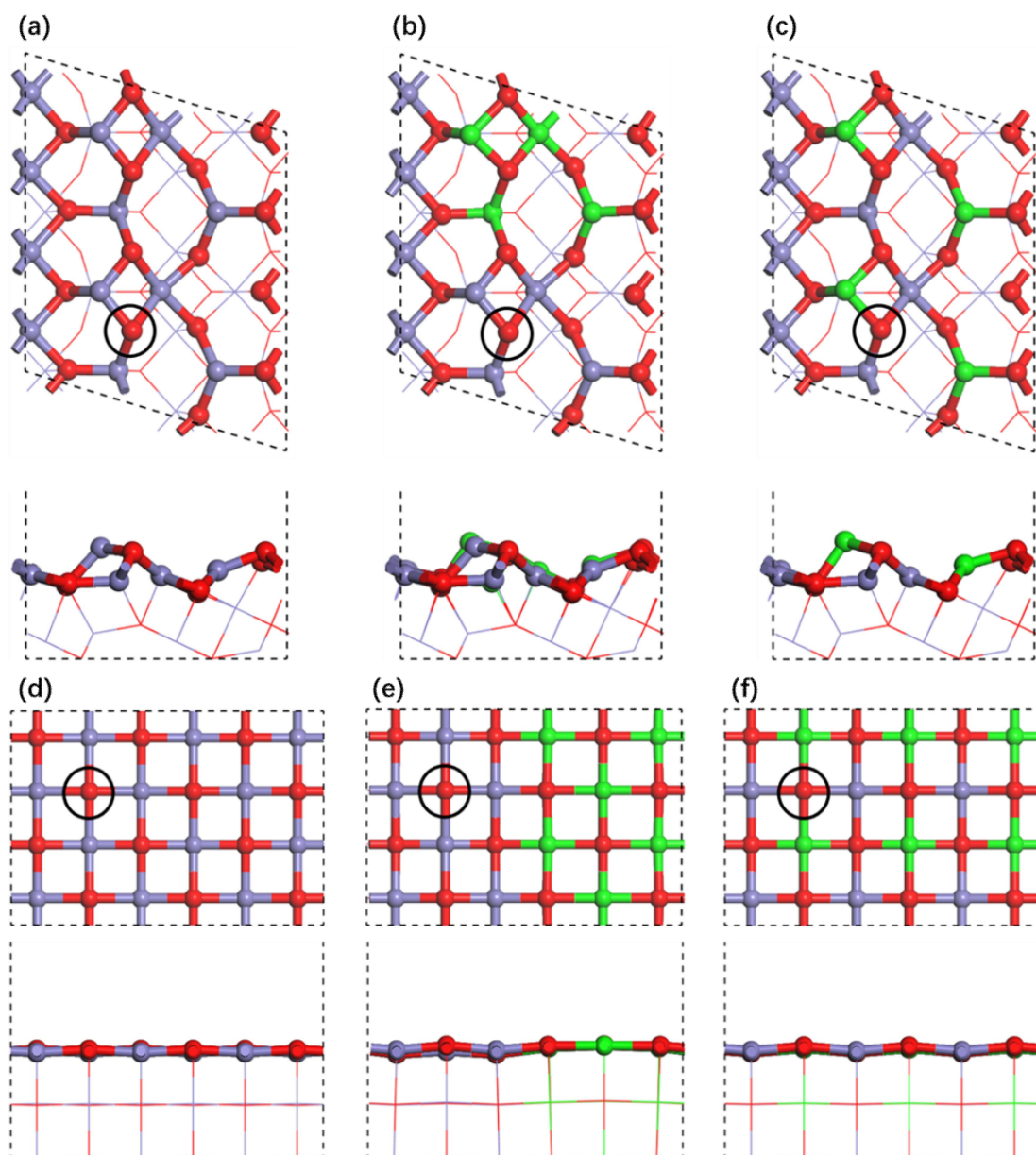


Fig S3. Top and side views of (a) Fe_3O_4 , (b) $\text{Fe}_3\text{O}_4_ \text{Mn}_3\text{O}_4$, (c) $(\text{Fe,Mn})_3\text{O}_4$ and (d) FeO , (e) FeO_MnO , (f) $(\text{Fe,Mn})\text{O}$ (Blue: Fe atoms; red: O atoms; green: Mn atoms). O atoms in the black circles are removed and substituted by C atoms.

Table S4. Detailed MES results of spent Fe4Mn1 catalysts.

	Hhf / KOe	IS / mm·s ⁻¹	QS / mm·s ⁻¹	Area	Phases
Fe4Mn1-CP	—	0.39	1.07	0.173	Fe ³⁺
	216.62	0.26	-0.12	0.302	
	185.68	0.2	0	0.349	Fe ₅ C ₂
	112.72	0.25	0.06	0.176	
	—	0.42	0.94	0.145	Fe ³⁺
Fe4Mn1-SCM	—	1.08	0.67	0.06	Fe ²⁺
	494.75	0.32	-0.01	0.07	
	460.83	0.54	0.07	0.153	Fe ₃ O ₄
	218.93	0.24	-0.11	0.208	
	192.78	0.2	0.01	0.205	Fe ₅ C ₂
	113.69	0.19	-0.09	0.107	
	172.99	0.19	-0.13	0.051	Fe _{2.2} C

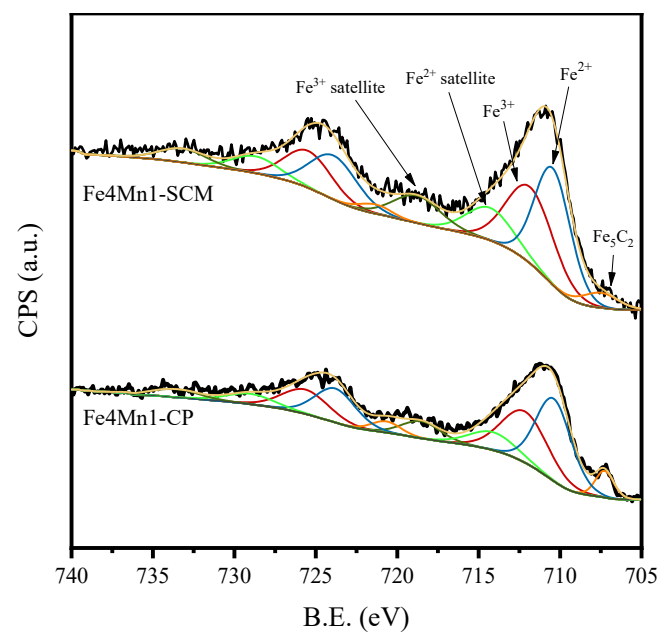


Fig S4. XPS spectra of spent Fe₄Mn₁ catalysts.

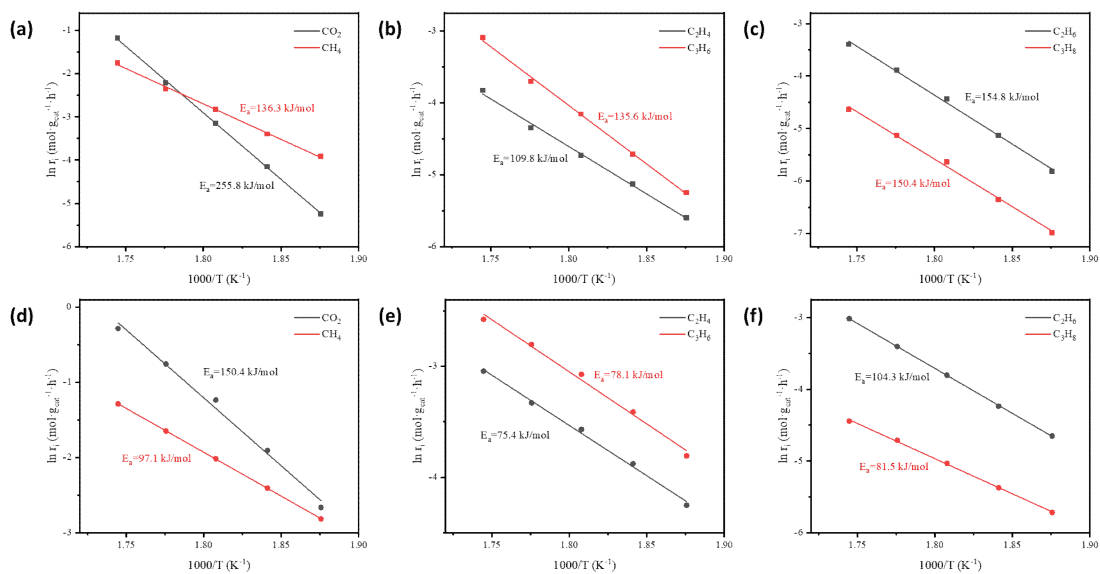


Fig S5. Activation energy (E_a) of C_1 - C_3 products for Fe4Mn1-SCM (a-c) and Fe4Mn1-CP (d-f). Kinetic measurements were taken at the following conditions: 260-300 °C, 2.0 MPa, 50ml/min, 15mg catalyst diluted with 85mg SiC

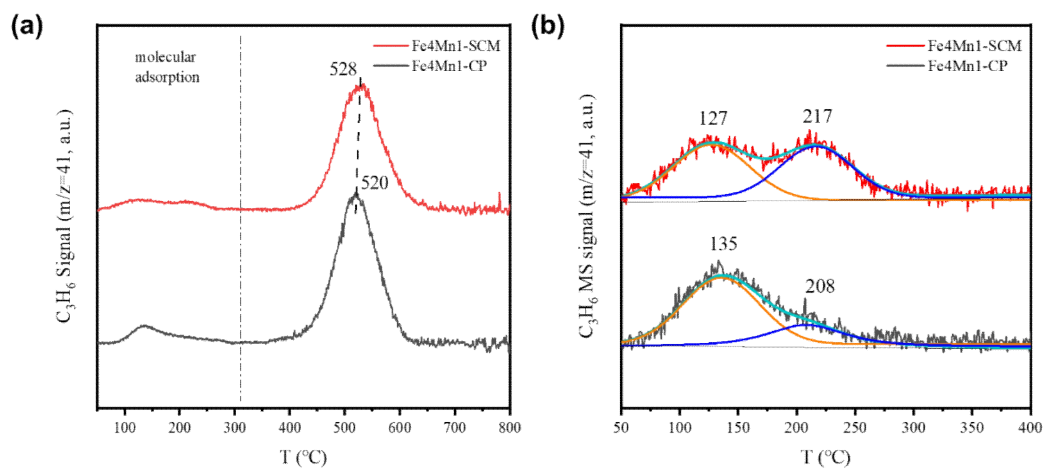


Fig S6. C_3H_6 -TPD profiles of the reduced Fe4Mn1 catalysts.

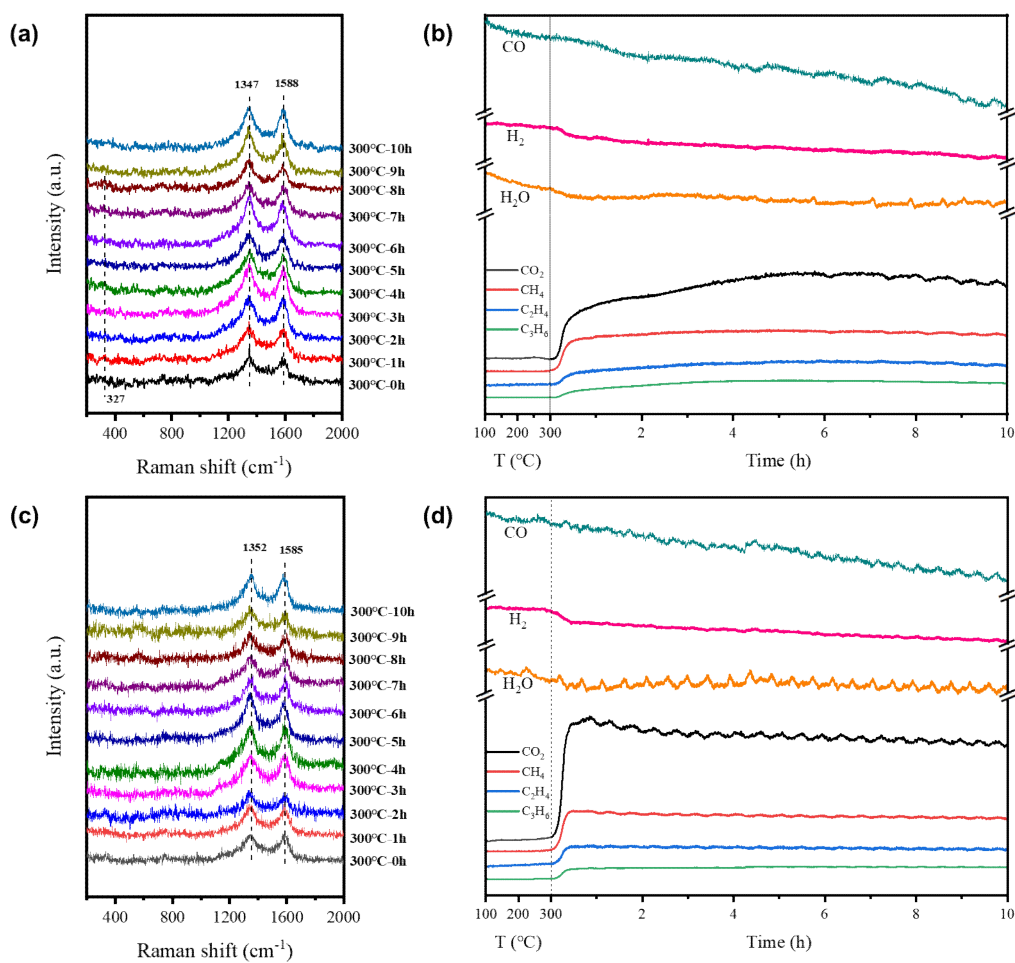


Fig S7. In-situ Raman-MS profiles of Fe₄Mn₁ catalysts during reaction process of Fe₄Mn₁-SCM (a, b) and Fe₄Mn₁-CP (c, d).

Table S5. Comparison of catalytic performance with the previous works.

Cat.	X _{CO} (%)	S _C O ₂ (%)	S _{HCS} (%)				FTY ($\mu\text{mol}_{\text{CO}} \cdot \text{g}_{\text{Fe}}^{-1} \cdot \text{s}^{-1}$)	Re f.
			C ₂₋ 4 ⁼	C ₂₋ 4 ⁰	C ₅₊	olefi ns		
MnO ₂ /α-Fe ₂ O ₃	48.2	29.4	51.1	8.9	26.6	-	181	[5]
Al-Fiber@meso-Al ₂ O ₃ @Fe-Mn-K	65.6	33.1	32.3	8.1	40.8	-	149	[6]
Fe-10MnK-AC	85	48	39.4	8.13	29.7	-	150	[7]
Fe/CeO ₂ @0.2MnO ₂	39.9	40.3	40.4	7.2	33.6	-	119	[8]
FeMn@SiO ₂ -0.25	67.2	47.5	20.9		71.8	-	-	[9]
FeMn/BN	19.8	7.5	43.4		37.0	-	7.86	[10]
FeMn10	83	34	26.6		63.9	-	23.2	[11]
0.5Mn/γ-Fe ₂ O ₃	63-68	36.8	58.4	9.5	21.2	-	23.1	[12]
FeMn-3.0Cu	96.9	23	40.1	30.4	9.5	-	5.39	[13]
6wt%Mn/Fe ₃ O ₄	41.5	37.8	60.1	6.5	23.7	-	16.8	[14]
FeMn@Si-c	56.1	13	-	-	-	75	16.8	[15]
Fe4Mn1-SCM	22.5	24.8	32.2	9.4	45.5	54.4	132	T W
Fe4Mn1-CP	32.	46.	51.	14.	15.	60.6	189	T

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“TW” is short for “this work”.

Table S6. Supplementary catalytic performance of different Fe and Fe4Mn1 catalysts^a

Catalysts	X _{CO} /%	S _{CO2} /	S _{HCS} (C-mol%, CO ₂ free)				O/P ^b	STY/g·kg _{cat} ⁻¹ ·h ⁻¹		Carbon balance/%
		%	CH ₄	C ₂₋₄ ⁰	C ₂₋₄ ⁼	C ₅₊		C ₂₋₄ ⁼	C ₂₋₇ ⁼	
Fe-SCM	21.0	35.0	21.7	21.2	40.5	16.7	1.91	1038.4	1322.8	97.7
Fe4Mn1-SCM	22.5	24.8	12.9	9.4	32.2	45.5	3.42	920.0	1148.8	97.6
Fe4Mn1-CP	32.4	42.8	19.8	14.0	51.2	15.1	3.65	1505.2	1782.0	98.4
Fe4Mn1-PM	34.9	44.8	22.2	18.4	42.1	17.3	2.28	1371.2	1748.5	97.1
Fe4Mn1Na-SCM	26.3	30.7	12.0	8.2	31.2	48.6	3.79	961.3	1173.2	97.0

^a Reaction conditions: 280 °C, 2.0 MPa, 45%CO/45%H₂/N₂, 7500 h⁻¹, TOS=24 h.

^b The ratio of olefins to paraffins of C₂-C₄.

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